

07_change_alignment

```
source("R/00_setup.R")
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
c2 <- readRDS("c2_matched.rds")
clsf <- readRDS("classification.rds") %>% select(country = iso, class)

#' ## Pass 1 - Net displacement
#'
#' `mag_ratio` below one means the model under-travels; `dir_cos` near one means
#' it moved the right way.

net <- c2 %>%
  group_by(model, country) %>%
  summarise(t1 = gt1[wave == max(wave)] - gt1[wave == min(wave)],
            t2 = gt2[wave == max(wave)] - gt2[wave == min(wave)],
            m1 = Dim1[wave == max(wave)] - Dim1[wave == min(wave)],
            m2 = Dim2[wave == max(wave)] - Dim2[wave == min(wave)],
            n_waves = n_distinct(wave), .groups = "drop") %>%
  mutate(T_mag = sqrt(t1^2 + t2^2),
         M_mag = sqrt(m1^2 + m2^2),
```

```

    mag_ratio = M_mag / T_mag,
    dir_cos = (t1 * m1 + t2 * m2) / (T_mag * M_mag),
    angle_deg = acos(pmax(pmin(dir_cos, 1), -1)) * 180 / pi) %>%
  left_join(clsf, by = "country")
saveRDS(net, "trajectory_net.rds")

moved <- net %>% filter(class != "stable")
stable <- net %>% filter(class == "stable")

cat("\n=== trajectory fidelity, moving countries ===\n")

```

=== trajectory fidelity, moving countries ===

```

moved %>% group_by(model) %>%
  summarise(n = n(),
    med_mag_ratio = median(mag_ratio), # ~1 tracks the rate, ~0 frozen
    med_dir_cos = median(dir_cos),
    frac_right_dir = mean(dir_cos > 0), # correct side of the map
    frac_within45 = mean(dir_cos > 0.707), # the tightened criterion
    .groups = "drop") %>%
  mutate(across(where(is.numeric), ~ round(.x, 3))) %>% print()

```

A tibble: 4 x 6

model	n	med_mag_ratio	med_dir_cos	frac_right_dir	frac_within45
<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1 Qwen/Qwen3.6-Plus	36	0.902	0.772	0.861	0.556
2 claude-opus-4-8	36	0.607	0.892	0.917	0.667
3 gemini-3.6-flash	36	0.817	0.871	0.833	0.694
4 gpt-5.5	36	0.672	0.708	0.861	0.5

```

#' The mirror check: where the survey records little movement, does the model
#' also stay put? Movement appearing where none should is the second failure
#' mode, and with four low-change countries this is a description rather than a
#' tested effect.

```

```

cat("\n=== low-change countries: model movement vs true movement ===\n")

```

=== low-change countries: model movement vs true movement ===

```
stable %>% group_by(model) %>%
  summarise(n = n(), med_model_move = median(M_mag), med_truth_move = median(T_mag),
    .groups = "drop") %>%
  mutate(across(where(is.numeric), ~ round(.x, 3))) %>% print()
```

```
# A tibble: 4 x 4
  model                n med_model_move med_truth_move
  <chr>              <dbl>         <dbl>         <dbl>
1 Qwen/Qwen3.6-Plus    4          0.145          0.097
2 claude-opus-4-8      4          0.126          0.097
3 gemini-3.6-flash     4          0.113          0.097
4 gpt-5.5              4          0.17           0.097
```

```
#' Rate flattening with intervals. Bootstrapped on the **median** because
#' `mag_ratio` has a near-zero denominator for reversals and a skewed
#' distribution. Run twice: all movers, and directional countries only, where net
#' displacement approximates path length so the ratio is well-behaved.
```

```
cat("\n=== median mag_ratio, all movers (upper CI < 1 = under-moves) ===\n")
```

```
=== median mag_ratio, all movers (upper CI < 1 = under-moves) ===
```

```
report_ci(boot_paired(moved, "mag_ratio", stat = median), "mag_ratio, all movers", vs = 1)
```

```
=== mag_ratio, all movers ===
                X2.5.  X50. X97.5. vs_ref
Qwen/Qwen3.6-Plus 0.698 0.902  1.032  n.s.
claude-opus-4-8   0.433 0.607  0.855  BELOW
gemini-3.6-flash  0.640 0.817  1.178  n.s.
gpt-5.5           0.548 0.690  0.971  BELOW
```

```
cat("\n=== sensitivity: directional countries only ===\n")
```

```
=== sensitivity: directional countries only ===
```

```
report_ci(boot_paired(net %>% filter(class == "directional"), "mag_ratio", stat = median),
  "mag_ratio, directional only", vs = 1)
```

```
=== mag_ratio, directional only ===
               X2.5.  X50.  X97.5. vs_ref
Qwen/Qwen3.6-Plus 0.437 0.684  0.792  BELOW
claude-opus-4-8   0.282 0.449  0.615  BELOW
gemini-3.6-flash  0.425 0.564  0.746  BELOW
gpt-5.5           0.346 0.576  0.776  BELOW
```

```
#' ## Pass 2 - Step-level alignment vs. a shuffled-year null

steps <- c2 %>% arrange(model, country, wave) %>%
  group_by(model, country) %>%
  mutate(ts1 = gt1 - lag(gt1), ts2 = gt2 - lag(gt2),
    ms1 = Dim1 - lag(Dim1), ms2 = Dim2 - lag(Dim2)) %>%
  filter(!is.na(ts1)) %>% ungroup()

step_cos <- function(d) {
  num <- d$ts1 * d$ms1 + d$ts2 * d$ms2
  den <- sqrt(d$ts1^2 + d$ts2^2) * sqrt(d$ms1^2 + d$ms2^2)
  mean(num / den, na.rm = TRUE)
}

#' The null keeps the model's steps and shuffles which true step they pair with.
#' Countries surveyed three times give two steps, which can be shuffled only by
#' swapping them, so per-country tests are underpowered by construction and
#' inference relies on the aggregate.

perm_null <- function(d, n = B) {
  k <- nrow(d); if (k < 2) return(NA_real_)
  replicate(n, {
    p <- sample(k)
    num <- d$ts1[p] * d$ms1 + d$ts2[p] * d$ms2
    den <- sqrt(d$ts1[p]^2 + d$ts2[p]^2) * sqrt(d$ms1^2 + d$ms2^2)
    mean(num / den, na.rm = TRUE)
  })
}

set.seed(SEED)
```

```

res <- steps %>% group_by(model, country) %>% group_split() %>%
  map_dfr(function(d) {
    obs <- step_cos(d); null <- perm_null(d)
    tibble(model = d$model[1], country = d$country[1], n_steps = nrow(d),
            obs_cos = obs, null_mean = mean(null, na.rm = TRUE),
            p_perm = mean(null >= obs, na.rm = TRUE))
  }) %>% left_join(clsf, by = "country")
saveRDS(res, "trajectory_stepcos.rds")

moved_steps <- res %>% filter(class != "stable")

cat("\n=== step-level alignment vs permutation null, moving countries ===\n")

```

=== step-level alignment vs permutation null, moving countries ===

```

moved_steps %>% group_by(model) %>%
  summarise(n = n(), med_obs_cos = median(obs_cos, na.rm = TRUE),
            med_null_cos = median(null_mean, na.rm = TRUE),
            frac_beats_null = mean(p_perm < 0.5, na.rm = TRUE),
            frac_sig = mean(p_perm < 0.1, na.rm = TRUE), .groups = "drop") %>%
  mutate(across(where(is.numeric), ~ round(.x, 3))) %>% print()

```

```

# A tibble: 4 x 6
  model          n med_obs_cos med_null_cos frac_beats_null frac_sig
  <chr>      <dbl>      <dbl>      <dbl>      <dbl>      <dbl>
1 Qwen/Qwen3.6-Plus 36      0.245      0.202      0.306      0
2 claude-opus-4-8 36      0.355      0.166      0.417      0
3 gemini-3.6-flash 36      0.454      0.244      0.556      0
4 gpt-5.5         36      0.175      0.171      0.333      0

```

```

#' ## Pass 3 - Drift-free alignment, bootstrapped
#'
#' The per-country gap between observed and null. This is the headline timing
#' result: an interval clearing zero is genuine tracking beyond the shared
#' modernization trend.

gaps <- res %>% filter(class != "stable") %>% mutate(gap = obs_cos - null_mean)
bg <- boot_paired(gaps, "gap")
report_ci(bg, "drift-free step alignment (obs - null), mean over moving countries")

```

```

=== drift-free step alignment (obs - null), mean over moving countries ===
      X2.5.  X50.  X97.5. vs_ref
Qwen/Qwen3.6-Plus -0.051 0.003  0.064  n.s.
claude-opus-4-8   0.001 0.083  0.173  ABOVE
gemini-3.6-flash  0.047 0.123  0.203  ABOVE
gpt-5.5           -0.019 0.049  0.114  n.s.

```

```
cat("\n=== raw observed vs null, point estimates ===\n")
```

```
=== raw observed vs null, point estimates ===
```

```

gaps %>% group_by(model) %>%
  summarise(mean_obs = round(mean(obs_cos, na.rm = TRUE), 3),
            mean_null = round(mean(null_mean, na.rm = TRUE), 3),
            mean_gap = round(mean(gap, na.rm = TRUE), 3), .groups = "drop") %>% print()

```

```

# A tibble: 4 x 4
  model          mean_obs mean_null mean_gap
  <chr>          <dbl>     <dbl>    <dbl>
1 Qwen/Qwen3.6-Plus 0.258     0.255    0.003
2 claude-opus-4-8   0.35      0.265    0.085
3 gemini-3.6-flash  0.404     0.281    0.123
4 gpt-5.5           0.288     0.24     0.049

```

```
report_pairwise(bg, "drift-free alignment")
```

```

=== drift-free alignment: pairwise difference (row - col), 95% CI; * = excludes 0 ===
      lo2.5 median hi97.5 sig
Qwen/Qwen3.6-Plus vs claude-opus-4-8 -0.184 -0.080  0.012
Qwen/Qwen3.6-Plus vs gemini-3.6-flash -0.207 -0.121 -0.036  *
Qwen/Qwen3.6-Plus vs gpt-5.5          -0.123 -0.047  0.046
claude-opus-4-8 vs gemini-3.6-flash -0.148 -0.039  0.074
claude-opus-4-8 vs gpt-5.5          -0.055  0.035  0.127
gemini-3.6-flash vs gpt-5.5          -0.014  0.074  0.159

```

```
#' ## Specification checks
#'
```

#' The test is blunt for three-wave countries, so it is cut two further ways:
 #' restricted to four-wave countries, where three steps make the permutation
 #' informative; and restricted to countries scored on a fuller indicator base.
 #' A model that clears zero in every specification is the only one whose timing
 #' signal is robust.

```
cat("\n=== four-wave countries only (3 steps) ===\n")
```

=== four-wave countries only (3 steps) ===

```
report_ci(boot_paired(gaps %>% filter(n_steps >= 3), "gap"), "drift-free alignment, 4-wave")
```

=== drift-free alignment, 4-wave ===

	X2.5.	X50.	X97.5.	vs_ref
Qwen/Qwen3.6-Plus	-0.060	0.028	0.114	n.s.
claude-opus-4-8	0.017	0.158	0.300	ABOVE
gemini-3.6-flash	0.066	0.204	0.338	ABOVE
gpt-5.5	0.012	0.101	0.200	ABOVE

```
k_items <- readRDS("model_coords.rds") %>%
  distinct(country, k_dim1, k_dim2) %>%
  mutate(k_tot = k_dim1 + k_dim2)

cat("\n=== dropping countries scored on fewer than 13 indicators ===\n")
```

=== dropping countries scored on fewer than 13 indicators ===

```
report_ci(boot_paired(gaps %>% inner_join(k_items, by = "country") %>% filter(k_tot >= 13),
  "drift-free alignment, fuller indicator base")
```

=== drift-free alignment, fuller indicator base ===

```
Warning in inner_join(., k_items, by = "country"): Detected an unexpected many-
to-many relationship between `x` and `y`.
i Row 1 of `x` matches multiple rows in `y`.
i Row 1 of `y` matches multiple rows in `x`.
i If a many-to-many relationship is expected, set `relationship =
  "many-to-many"` to silence this warning.
```

	X2.5.	X50.	X97.5.	vs_ref
Qwen/Qwen3.6-Plus	-0.036	0.020	0.079	n.s.
claude-opus-4-8	-0.032	0.063	0.155	n.s.
gemini-3.6-flash	0.008	0.096	0.189	ABOVE
gpt-5.5	-0.050	0.029	0.102	n.s.